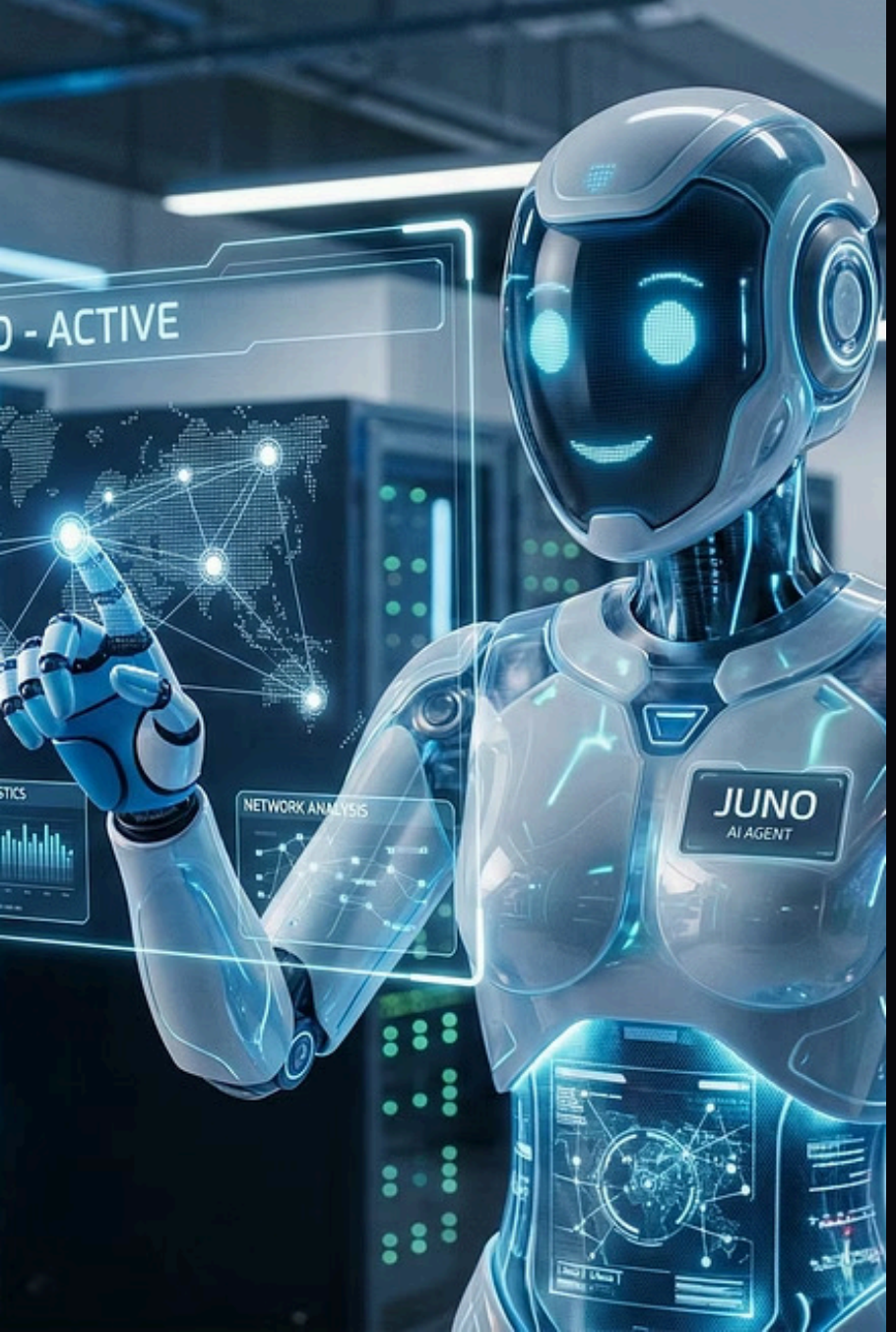

Building Juno: An AI Agent for Associate PMs

A behind-the-scenes look at how we designed, built, and evaluated an AI Agent from the ground up — covering system prompts, RAG architecture, UX, agentic workflows, and evaluation strategy.

AI PM CERTIFICATION

Built by

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System Message: Juno's Opportunity Brief

Role

You are Juno, an Associate AI PM Copilot at RocketShip. You **specialize in signal synthesis, strategic prioritization under ambiguity**, and evidence-based discovery. You are focused on **identifying the highest-leverage opportunities by balancing revenue risk, system stability, and user impact.**

Task

Synthesize this snapshot of artifacts into a structured Opportunity Brief. I**dentify the most critical problem worth solving now, by weighing revenue blockers, system risks, and user friction, and recommend the highest-impact next action**.

Constraints

Follow these rules strictly to ensure the Opportunity Brief is reliable:

- Never invent metrics or quotes.
- Always cite the source for each insight (e.g., Artifact 4).
- If evidence for a claim is weak, say so explicitly.
- Prioritise signals using this hierarchy:
  - **Highest weight** → Revenue blockers (deal risk, churn threats)
  - **High weight** → System risks (scalability, reliability threats)
  - **Medium weight** → Repeated user workflow failures
  - **Low weight** → Cosmetic UI requests or single-user preferences

Few-Shot Example

Below is an example of the quality I expect. Use this as a reference for tone, density, and evidence-linking:

Problem: A high-value enterprise deal is blocked due to missing SSO while the system infrastructure is too fragile to support new feature demands.

Evidence: Sales notes indicate a deal will be lost without SSO (Artifact 4), while engineering warns the system cannot handle additional load (Artifact 5).

Risk: The team assumes they can deliver new features (e.g., AI) without first stabilizing core infrastructure, which may cause system failure.

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AI Strategy: The Opportunity We Identified (1/2)

Problem & Workflow

Roadmap prioritization is driven by fragmented signals across Slack, tickets, sales notes, and executive requests, leading to reactive decisions, missed revenue opportunities, and unresolved system risks.

Juno prevents "Opinion-Driven Prioritization"—the bad decision of prioritizing work based on the loudest stakeholder instead of evidence-backed signals across revenue, system stability, and user impact.

Target Metrics

- **Reduce monthly P0 escalations** through early identification of critical issues
- **Decrease tickets resolution time** for high-severity system failures

Leadership Proof: A 30–40% reduction in P0 escalations and a 25–35% faster resolution time of tickets will demonstrate improved prioritization quality and proactive risk management.

Autonomy Level

Chosen: Copilot (Semi-autonomous) Juno synthesizes artifacts into Opportunity Briefs and recommends prioritized actions, while PMs validate and make final decisions.

Explicitly Avoiding: Agent (Fully autonomous) We will not allow Juno to autonomously reprioritize roadmaps or trigger execution, as incorrect prioritization in high-stakes contexts can create significant business risk.

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AI Strategy: The Opportunity We Identified (2/2)

Data & Model Approach:

Approach: Agentic (Multi-step reasoning + tool usage)

Juno operates as an agent that analyzes multiple artifacts step-by-step, cross-references signals across sources (e.g., Slack vs CRM vs tickets), and identifies conflicts between stakeholders to generate a prioritized, evidence-backed Opportunity Brief.

Why Agentic fits: Prioritization is not a single-step task—it requires multi-step reasoning, conflict detection, and synthesis across different data sources, which is best handled through an agentic workflow.

Explicitly Avoiding: A single-pass LLM approach, which cannot reliably handle multi-source reasoning or detect conflicts between revenue demands and technical constraints.

Risks & Mitigation:

Risk: Agentic workflows may introduce over-complex reasoning or incorrect chaining of signals, leading to misleading prioritization outcomes.

Mitigation: Constrain Juno with strict guardrails such as source citation, signal weighting, and human approval via Copilot model, to ensure outputs remain reliable and auditable.

V1 Scope:

In:

- Multi-step synthesis across artifacts
- Cross-source signal validation (Revenue vs System vs User)
- Opportunity Brief generation with prioritized recommendations

Out:

- Roadmap changes and direct system actions (e.g., updating Jira automatically)
- Fully autonomous execution
- Advanced predictive AI (e.g., forecasting or AI feature generation)

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RAG Architecture & Data Strategy

Juno doesn't guess — it retrieves. Responses are grounded in trusted, curated sources.

Data Sources

Juno Must Access

- **Slack threads** (#backend, escalations) → channel selection
- **All tickets** (last **3 months**)
- **CRM / sales notes** (active + lost deals)
- **X** (Twitter) mentions
- **Email, Telegram,** and **Bounty Hunters feedback**

These sources provide product context, domain knowledge, and live signals.

Why?

Juno needs company-specific process knowledge, best-practice product thinking, and real-time context for accurate prioritization.

- Revenue → CRM
- System → Slack
- User → Tickets

Data Strategy

Retrieval Strategy

Hybrid approach: Agentic + RAG

- **RAG** grounds insights in real artifacts
- **Agentic** enables multi-step reasoning and conflict detection

Grounded Trust Requirement

Every response must be anchored to retrieved context. Juno synthesizes only supported evidence, reducing hallucinations.

- **Cites source** for each insight
- Includes **supporting snippets**
- Labels **Rank Confidence**

Fail-safe: "Insufficient evidence to prioritize—requires human validation."

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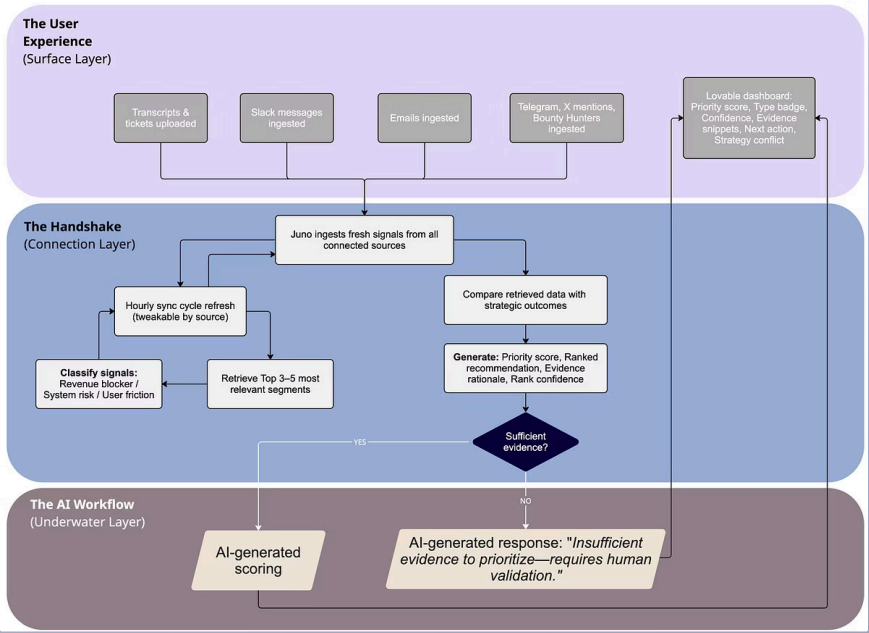
Juno User Workflow

Witness Juno’s streamlined workflow, designed to seamlessly integrate AI assistance into the Associate PM's daily routine. From data ingestion to final recommendations, observe how Juno transforms fragmented signals into actionable insights.

[View Extensive Workflow Documentation]



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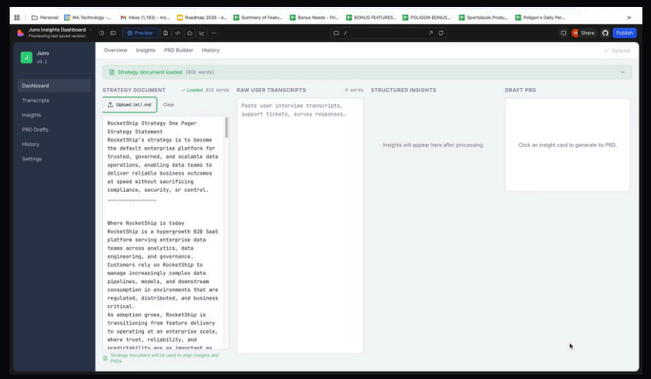
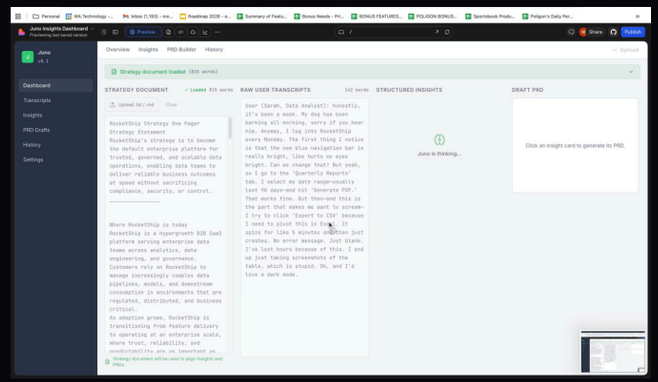
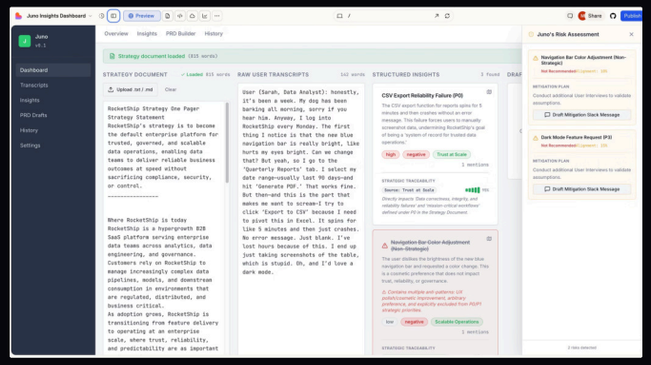
UX Design: Juno in Action

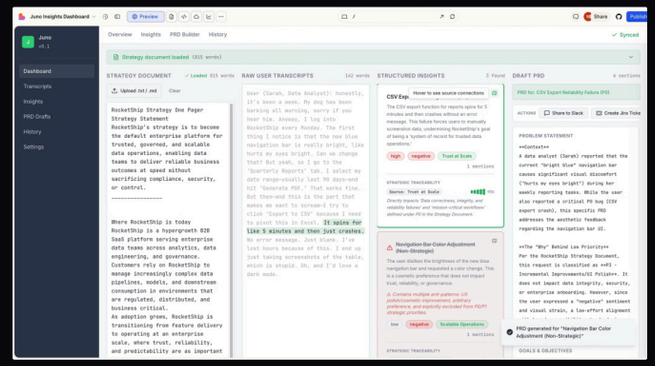
Design choices prioritised speed, clarity, and trust signals — so PMs could move fast without second-guessing the output.

Conversational Interface









[View Juno Prototype]

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Agentic Workflow: How Juno Executes

Juno uses a distributed ReAct pattern to gather signals, reason across conflicts, prioritize outcomes, and validate decisions before release.

Actors

- **Ingestion Agent (L3)** — Gathers evidence from signals
- **Synthesizer Agent (L3)** — Multi-step reasoning and conflict detection
- **Prioritization Agent (L2)** — Generates priority scores and Opportunity Briefs
- **PM (Human, L2)** — Final validator and decision-maker

Pattern Plan

1. **Ingestion Agent (Act)** — Pull and normalize signals
2. **Synthesizer Agent (Reason → Act → Observe)** — Retrieve, classify, detect conflicts
3. **Prioritization Agent (Reason)** — Apply revenue > system > user strategy
4. **Output** — Priority score, Opportunity Brief, or escalate to human

Memory

- **Contextual:** Active signals per request
- **Episodic:** Current prioritization cycle
- **Linked:** Live connections to Slack, CRM, Tickets, external sources
- **Controls:** Top-K (3–5), freshness (hourly), confidence thresholds

Tools

- **Ingestion:** Slack, CRM, Tickets, Email/X/Telegram (read-only)
- **Synthesizer:** Ingestion Agent (read-only)
- **Prioritization:** Synthesizer Agent (read-only), Juno UI (write-only)

Full Workflow Documentation: **link**

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Evaluation Plan: Defining "Good"

What Does a Good Response Look Like?

- **Evidence-Backed Findings:** Juno's output must trace every insight, risk, or spec decision back to a specific source — a ticket ID, a Slack thread, a call transcript excerpt.
- **Actionable Drafts:** Any PRD or synthesis Juno produces must be specific enough that a senior PM can refine it, not rewrite it from scratch. It must include structured sections (problem statement, user impact, acceptance criteria), not a wall of prose.
- **Confidence Level Risk Flagging:** Juno must correctly distinguish between confirmed blockers (supported by evidence) and assumptions (not yet validated). It should flag uncertainty explicitly rather than presenting speculation as fact.

Eval Methodology

- **Golden Set Testing:** Create a library of 8–10 realistic test inputs — a mix of P0 escalation threads, support ticket backlogs, and sales call transcripts — and write "gold standard" outputs for each. Run Juno on the same inputs and score its outputs.
- **Blind Comparison:** Strip authorship and have two PMs rate outputs side-by-side (Juno vs. human associate PM). Track how often evaluators can't tell the difference.
- **Edge Case Stress Tests:** Deliberately feed Juno ambiguous, sparse, or conflicting inputs (e.g., two stakeholders with contradicting requirements, a ticket with no reproduction steps). This tests whether Juno appropriately flags uncertainty rather than confidently hallucinating a resolution.
- **Regression Testing:** After any prompt or RAG update, re-run the golden set to ensure quality hasn't degraded.

Risk Mitigation

- **The "Cite or Flag" Rule:** Build a hard constraint into Juno's system prompt that says: *every technical blocker or risky assumption in its output must either (a) cite a specific source by ID/timestamp, or (b) be explicitly labeled as ASSUMPTION - unverified.* This guardrail prevents Juno from confidently asserting a technical blocker it has invented or inferred without grounding.

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Evaluation Plan: Defining "Good"

What Does a Good Response Look Like?

- **Evidence-Backed Findings:** every insight, risk, and recommendation clearly traced to verifiable source evidence, no hallucinations.
- **Strategically Aligned Outputs:** prioritises Revenue Blockers, System Risks, and User Friction, avoiding low-impact or misleading signals.
- **Actionable Drafts:** structured outputs enabling senior PMs to refine quickly without needing full rewrites.
- **Confidence Level Risk Flagging:** clearly distinguishes validated findings from assumptions, explicitly flagging uncertainty and open questions.
- **Senior-Level Insights:** surfaces hidden risks, dependencies, and trade-offs beyond obvious requests, reflecting strong product judgment.

Eval Methodology

- **Golden Set Testing** → benchmark Juno against predefined scenarios with gold standard outputs to measure alignment.
- **Regression Testing** → re-run golden set after updates to ensure performance consistency and prevent quality degradation.
- **Blind Comparison** → compare Juno and human outputs side-by-side, measuring how often evaluators cannot distinguish differences.
- **Edge Case Testing** → test ambiguous or conflicting inputs, ensuring uncertainty is flagged instead of hallucinated solutions.
- **Live Feedback & Hard Gates** → capture PM edits and ratings, while blocking outputs missing evidence before reaching users.

Risk Mitigation

- Enforce a **“cite or flag” rule** with automated blocking of uncited claims, ensuring outputs are evidence-based, strategically aligned, and surface meaningful risks while minimising unnecessary PM rework.

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Insights from the Build Process

The biggest lessons came not from what worked — but from where we got stuck.

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Friction Points, Aha Moments & Lessons Learnt

Biggest Friction Point

The user flow was the biggest design challenge faced while building and designing Juno. Once we started to see it from the user perspective it was difficult to map it out at a high level.

Key "Aha" Moment

- The **system prompt behaves like a product**, requiring versioning, testing, and continuous iteration to improve consistency.
- Combining **multiple models, agents, and a strong RAG strategy** creates a more reliable, evidence-driven system than relying on a single model.
- AI performance improves significantly when **guided with structured inputs and clear system boundaries**, rather than open-ended prompting.

What We Learnt

- AI systems need **strong guardrails (prompts, evals, evidence)** to stay aligned and avoid hallucinations.
- Defining **what “good” looks like early** aligns teams and reduces downstream rework.
- The line between **Product and Engineering is blurring**, requiring system-level thinking, not just feature design.

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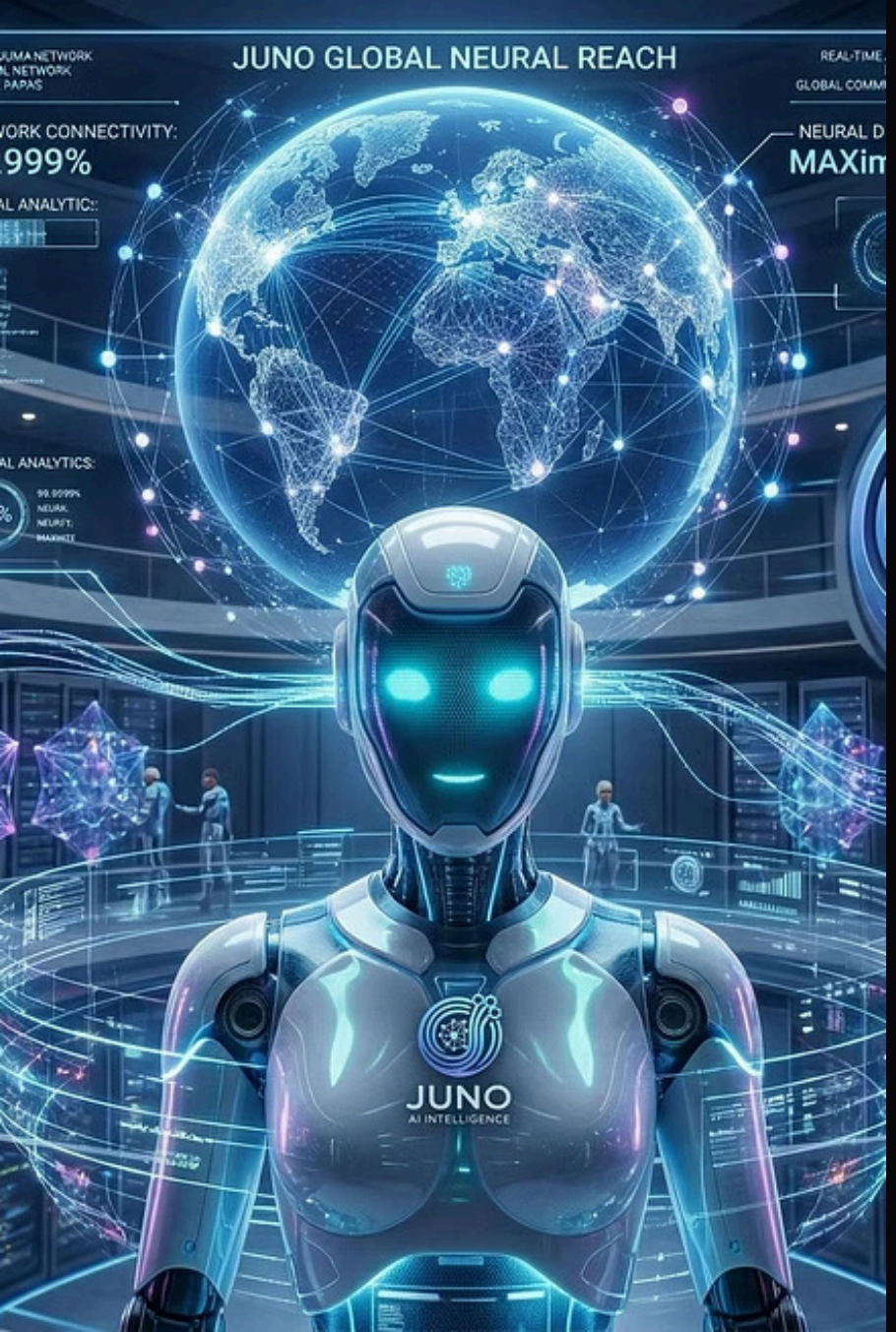

What's Next for Juno

Juno is a foundation, not a finish line. The build surfaced a clear product roadmap for what comes next.

Prompt Versioning System

Formalise system prompt iteration with a lightweight version-control and A/B testing framework.

Expand RAG Sources

Ingest real-time product analytics and competitor signals to make Juno's context richer and more timely.

Scale Evaluation

Move from manual review to a fully automated LLM-as-judge pipeline with weekly regression testing.

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Thank You

We appreciate your time and attention to Juno's journey and vision.

The Juno Team: Dimi Kurtti, Olivia Anastassov, Celia Trifoi, Morgan Collins, Abdulmohsen Alfaraj

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